In [1]:
```
# ==========================================
# E-Commerce Data Analysis Project
# Author: Rui Cristovam
# Description: SQL + Python analysis using SQLAlchemy
# ==========================================
```

In [2]:
```
!pip install pymysql
```

```
Requirement already satisfied: pymysql in c:\users\alvar\anaconda3\lib\site-packages (1.1.2)
```

In [3]:
```
import pandas as pd
from sqlalchemy import create_engine

# 1. Credentials Configuration
user = "root"
password = ""  # Se tiveres password, coloca-a aqui
host = "localhost"
database = "ecommerce_project"

# 2. Create SQLAlchemy Engine
# The 'mysql+pymysql' prefix is the recommended standard for MySQL connections
engine = create_engine(f"mysql+pymysql://{user}:{password}@{host}/{database}")

print("Ligação via SQLAlchemy: OK")

# 3. SQL Query for Data Extraction
query = """
SELECT
    o.order_purchase_timestamp,
    p.payment_value,
    c.customer_state
FROM orders o
JOIN order_payments p ON o.order_id = p.order_id
JOIN customers c ON o.customer_id = c.customer_id
"""

# 4. Load Data using the engine
df = pd.read_sql(query, engine)

# 5. Data Preprocessing: Convert timestamp to datetime objects
df['order_purchase_timestamp'] = pd.to_datetime(df['order_purchase_timestamp'])

# Displaying initial results
print(f"Dados carregados: {len(df)} linhas.")
df.head()
```

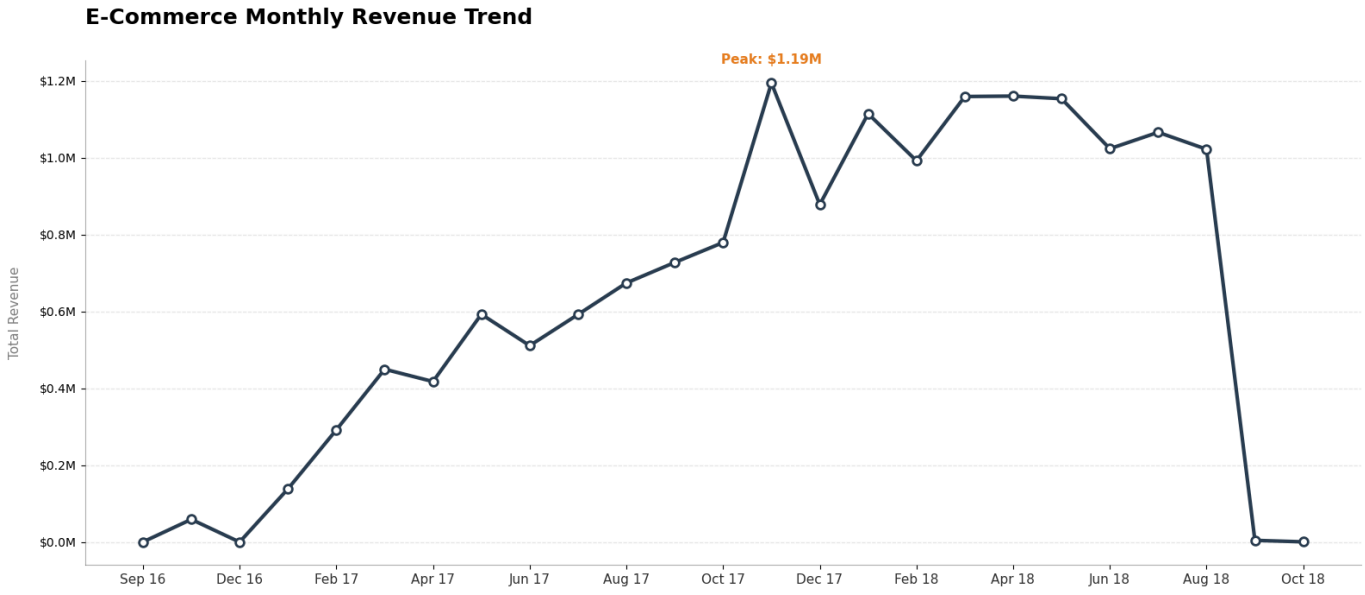
```
Ligação via SQLAlchemy: OK
Dados carregados: 103886 linhas.
```

Out[3]:

	order_purchase_timestamp	payment_value	customer_state
0	2017-07-29 20:08:40	342.56	MG
1	2017-03-10 20:04:32	130.22	SP
2	2017-04-23 18:13:00	208.41	SP
3	2018-03-07 15:32:34	67.42	SP
4	2017-02-20 19:21:23	232.05	RJ

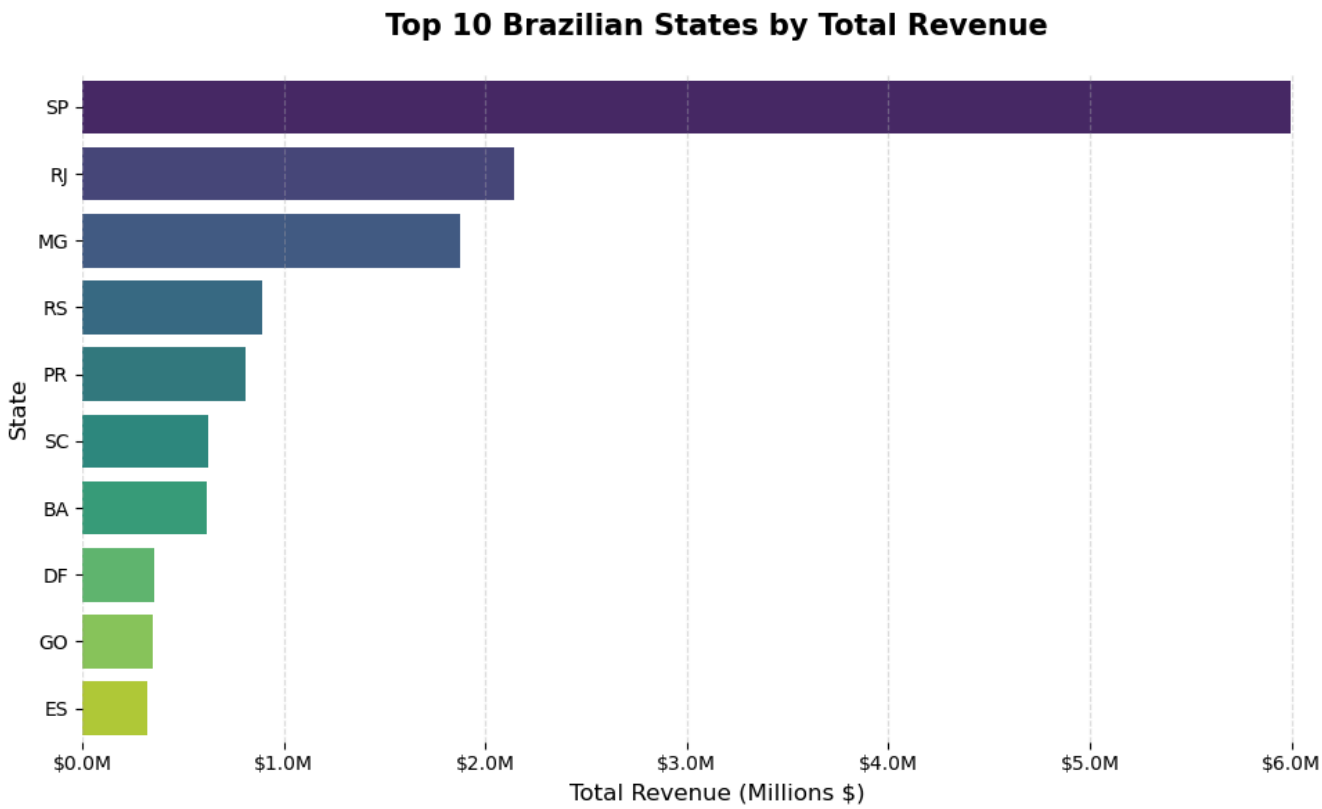
In [4]:
```
import matplotlib.pyplot as plt
import seaborn as sns
import matplotlib.ticker as ticker
```

```python
# --- 1. Data Preparation ---
# Sort by date and create a monthly period column
df_sorted = df.sort_values('order_purchase_timestamp')
df_sorted['month_year_dt'] = df_sorted['order_purchase_timestamp'].dt.to_period('M')

# Aggregate total revenue per month
monthly_revenue = df_sorted.groupby('month_year_dt')['payment_value'].sum().reset_index()

# Formatting labels for the X-axis (e.g., "Jan 17")
monthly_revenue['month_label'] = monthly_revenue['month_year_dt'].dt.strftime('%b %y')

# --- 2. Chart Creation ---
plt.figure(figsize=(16, 7), facecolor='white')
ax = plt.gca()

plt.plot(monthly_revenue['month_label'], monthly_revenue['payment_value'],
         color='#2c3e50', linewidth=3, marker='o', markersize=7,
         markerfacecolor='white', markeredgewidth=2)

# --- 3. X-Axis Strategy (Optimizing readability) ---
# Displaying only 1 out of every 2 months to provide more visual space
ticks_to_show = range(0, len(monthly_revenue), 2)
labels_to_show = [monthly_revenue['month_label'][i] for i in ticks_to_show]

plt.xticks(ticks_to_show, labels_to_show, fontsize=11, color='#333333', rotation=0)

# --- 4. Aesthetics & Formatting ---
plt.title('E-Commerce Monthly Revenue Trend', fontsize=18, pad=30, loc='left', fontweight='bo
plt.ylabel('Total Revenue', fontsize=11, color='gray', labelpad=15)

# Y-Axis Formatting (Transforming values to Millions $)
formatter = ticker.FuncFormatter(lambda x, pos: f'${x/1e6:.1f}M')
ax.yaxis.set_major_formatter(formatter)
plt.yticks(fontsize=10)

# Visual Cleanup: Removing borders and adding subtle grids
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)
ax.spines['left'].set_alpha(0.3)
ax.spines['bottom'].set_alpha(0.3)
plt.grid(axis='y', linestyle='--', alpha=0.3)

# Highlight: Peak Value Annotation
max_val = monthly_revenue['payment_value'].max()
max_idx = monthly_revenue['payment_value'].idxmax()
plt.text(max_idx, max_val + 50000, f'Peak: ${max_val/1e6:.2f}M',
         ha='center', fontweight='bold', color='#e67e22', fontsize=11)

plt.tight_layout()
plt.show()
```



In [5]:

```
import matplotlib.pyplot as plt
import seaborn as sns
import matplotlib.ticker as ticker

# 1. Data Preparation: Group by state and calculate top 10 by total revenue
state_revenue = df.groupby('customer_state')['payment_value'].sum().sort_values(ascending=Fal

# 2. Chart Configuration
plt.figure(figsize=(10, 6))

# Utilizing 'hue' and 'legend=False' to ensure compatibility with modern Seaborn versions
ax = sns.barplot(
    x=state_revenue.values,
    y=state_revenue.index,
    hue=state_revenue.index,
    palette='viridis',
    legend=False
)

# 3. Aesthetics & Titles
plt.title('Top 10 Brazilian States by Total Revenue', fontsize=15, pad=20, fontweight='bold')
plt.xlabel('Total Revenue (Millions $)', fontsize=12)
plt.ylabel('State', fontsize=12)

# 4. X-Axis Formatting: Convert large numbers to Millions ($M) format
formatter = ticker.FuncFormatter(lambda x, pos: f'${x/1e6:.1f}M')
ax.xaxis.set_major_formatter(formatter)

# Visual Cleanup
sns.despine(left=True, bottom=True)
plt.grid(axis='x', linestyle='--', alpha=0.4)

plt.tight_layout()
plt.show()
```



📈 Strategic Business Insights: E-Commerce Performance

1. Revenue Dynamics & Seasonality

- **Black Friday Catalyst:** The massive spike in **November 2017** confirms that the operation is highly reactive to major promotional events. This suggests a need for robust inventory planning 3 months prior to Q4.
- **Consistent Growth:** Beyond seasonal peaks, the month-over-month uptrend (2017-2018) shows healthy market expansion and increasing customer lifetime value.

2. Geographical Dominance & Expansion

- **The Southeast Hub: São Paulo (SP)** is the primary revenue driver. While this shows strength, it also represents a regional dependency risk.
- **Strategic Recommendation:** Optimize logistics and "Last Mile" delivery in SP, RJ, and MG to protect margins, while piloting targeted acquisition campaigns in Southern states to diversify the revenue base.

3. Hourly Sales Velocity

- **Prime Time Windows:** Orders peak between **10:00–12:00** and **19:00–21:00**.
- **Tactical Application:** Marketing spend (ads, emails, push notifications) should be concentrated 30 to 60 minutes before these peak windows to maximize the Conversion Rate (CR).

Executive Summary: The business is in a strong growth phase. Future success depends on maintaining the momentum in the Southeast while leveraging data-driven marketing timing to optimize the return on advertising spend (ROAS).